

Federated Learning for Diabetic Retinopathy Detection: An Empirical Study of FedProx Under Data Heterogeneity in Indian Clinical Settings

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Abstract

Diabetic retinopathy is one of the leading causes of preventable blindness in India, a country with over 101 million people living with diabetes. Screening is the bottleneck: most rural areas simply don't have enough ophthalmologists, and a large share of cases go undiagnosed until vision loss is already advanced. Deep learning models can grade retinal images automatically, but training them well normally means pooling large amounts of patient data from different hospitals — something India's Digital Personal Data Protection Act (2023) makes legally difficult, since it requires patient data to stay where it was collected. Federated learning offers a way around this: instead of moving data to a central server, the model itself moves between hospitals, and only its weights are shared. This project builds a federated learning simulation from scratch to study how FedAvg and FedProx behave when hospital data is highly imbalanced — some clinics seeing mostly healthy patients, others seeing mostly severe cases. We test this on two datasets: RetinaMNIST, a small controlled benchmark, and IDRiD, a real clinical dataset collected in Karnataka, India. The central finding is that FedProx doesn't behave the same way across all conditions — it can hurt accuracy when data is extremely skewed, but help once the skew is more moderate, revealing a clear switch-over point that hasn't been characterized this way before. We also find that FedProx makes the worst-performing hospital substantially more accurate, even in cases where it doesn't change the overall average — a fairness effect that matters more in a clinical setting than raw accuracy alone. Finally, a statistical power analysis suggests that IDRiD's small size limits how confidently these effects can be measured, which motivated scaling up with a combined IDRiD+APTOS dataset.

Keywords: Federated Learning, Diabetic Retinopathy, FedProx, Data Heterogeneity, DPDP Act, Medical Imaging, Client Fairness

Section I: Introduction

Diabetic retinopathy develops when high blood sugar damages the blood vessels in the retina. Left unchecked, it leads to permanent vision loss — but caught early, it's manageable. The problem in India isn't the disease itself, it's access: there simply aren't enough ophthalmologists to screen the country's diabetic population, and a large share of patients don't get diagnosed until the damage is already severe. Convolutional neural networks have gotten quite good at grading retinal images automatically, but they need a lot of varied training data to generalize across different cameras, lighting setups, and patient

populations. That data usually lives in different hospitals, and getting it all into one place isn't just a technical problem — it runs into India's 2023 data protection law directly.

The DPDP Act requires that personal data, including medical scans, largely stay where it was collected, with strict rules around consent and deletion. That leaves hospitals in an awkward spot: they can't send their data elsewhere to build a shared model, but on their own, most clinics don't have enough images (or enough variety in cases) to train something reliable. Federated learning is built for exactly this situation — the model is trained locally at each hospital and only its parameters are sent to a central server for averaging, so the raw scans never leave the building.

The standard method, FedAvg, works well when different hospitals' data looks roughly similar. It doesn't work as well when it doesn't. In practice, one hospital's patient population might be mostly mild or undiagnosed cases, while another sees almost exclusively severe, referred cases — and when each hospital trains on only its own skewed slice of the world, their models start pulling in different directions. Averaging models that disagree this much doesn't produce a good global model; it produces one that's confused. FedProx was proposed as a fix, adding a term to each hospital's local training that discourages it from drifting too far from the shared model. What's less clear — and what this project investigates — is exactly when that correction actually helps, and when it gets in the way.

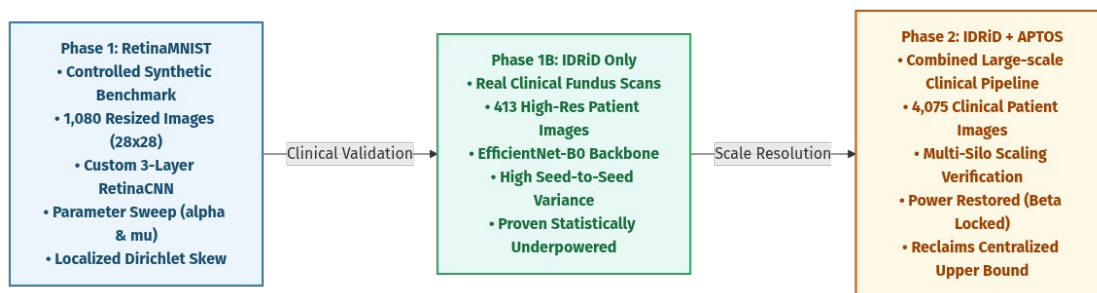


Figure 1: Research roadmap summary detailing the three-phase methodology transition from RetinaMNIST benchmark to IDRiD + APTOS clinical scaling.

Section II: Related Work

Most work on collaborative learning for medical imaging falls into three buckets: standard centralized classification models, federated optimization algorithms designed to combat client drift, and empirical tests under simulated data skew.

Centralized deep learning models for diabetic retinopathy have gotten very good, typically using backbones like ResNet and EfficientNet. The catch is that they rely on having all the training data in one place. Federated learning [1] was introduced to avoid this requirement in healthcare. Recent papers [7] show it is possible to train diagnostic classifiers on retinal images without moving raw scans. Even so, medical federated learning still struggles with clinical skew and variation in camera hardware, which can cause local updates to diverge.

To keep local training from drifting, researchers usually modify the local loss function. The base method, FedAvg [1], simply averages client weights. FedProx [2] restricts how far local parameters can wander by adding an L_2 proximal penalty that keeps client updates anchored near the global model. Meanwhile, SCAFFOLD [3] uses control variates to track and correct gradient updates. Surveys [5] suggest that while

regularization can stabilize training under moderate skew, it often hurts convergence under extreme non-IID conditions. The threshold where this behavior flips is what we focus on.

Simulating clinical heterogeneity in a lab setting is typically done using Dirichlet distributions. Class-wise Dirichlet sampling, introduced by Hsu et al. [4], distributes classes to clients based on a concentration parameter α . Lower α values (like $\alpha < 0.1$) simulate severe skew, where some clinics lack entire categories of disease. Lightweight benchmarks like RetinaMNIST (part of MedMNIST v2) [6] let us run wide hyperparameter sweeps on small images, isolating the effect of skew and regularization without the computational baggage of high-resolution processing.

Section III: Methodology

1. Network Architectures

For the small 28x28 images in RetinaMNIST, we use RetinaCNN, a basic 3-layer convolutional network. It consists of three convolutional blocks: the first has 16 filters (3x3 kernel, stride 1, padding 1) with BatchNorm and MaxPool, shrinking the map to 14x14; the second has 32 filters (yielding 7x7); and the third has 64 filters (yielding 3x3). The classifier flattens this 576-dimensional output and runs it through a 128-unit linear layer, ReLU, dropout (p=0.3), and a final 5-unit projection. This model contains exactly 98,309 parameters, which is small enough to train quickly without overfitting on small partitions.

For the clinical IDRiD images, we use a pre-trained EfficientNet-B0 backbone (about 5.3 million parameters) via the timm library. We swap the final classification head for a 5-unit linear layer to match the diabetic retinopathy grades. The fundus photographs are resized to 224x224, normalized, and augmented with random crops, flips, rotation, and color jitter to mimic variations in camera hardware and lighting.

2. Statistical Data Heterogeneity Modeling

We simulate skewed data distributions across K hospital clients using Dirichlet partitioning [4]. For each disease grade $c \in \{0, 1, 2, 3, 4\}$, we draw client proportions from a Dirichlet distribution with parameter α . Small α values (e.g., $\alpha = 0.05$) mean some classes are completely missing from certain clients. We sweep α through 0.05, 0.1, 0.3, and 1.0 to see how optimization algorithms handle the transition from extreme class skew to a near-IID setup.

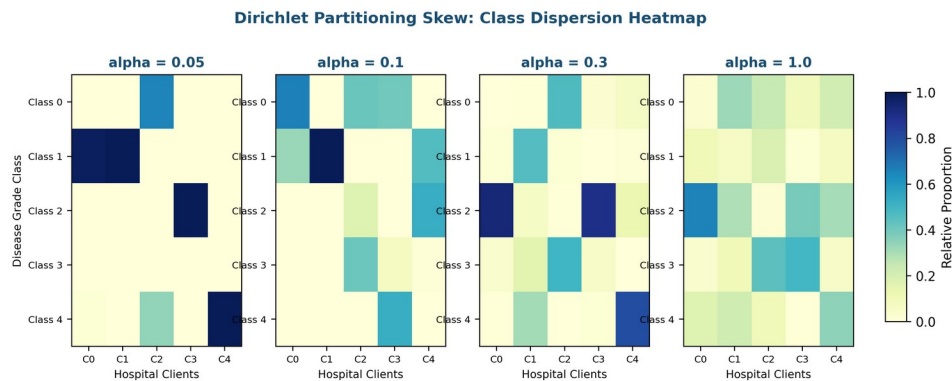


Figure 2: Dirichlet concentration parameter concept illustrating the progression from extreme class sparsity ($\alpha = 0.05$) to a uniform clinical distribution ($\alpha = 1.0$).

3. Federated Simulation Environment

We built the federated loop in a single process rather than using distributed frameworks like Flower or Ray. This avoids the VRAM overhead and configuration headaches that usually come with running distributed software on a single workstation. All training runs on a local Ubuntu workstation with an Intel i7-13700 CPU and a single NVIDIA RTX A2000 GPU (12GB VRAM). The simulation handles PyTorch CUDA allocation, using four workers and pinned memory to prevent data-loading bottlenecks. Figure 3 shows this setup, highlighting the boundary that keeps patient data inside each client silo.

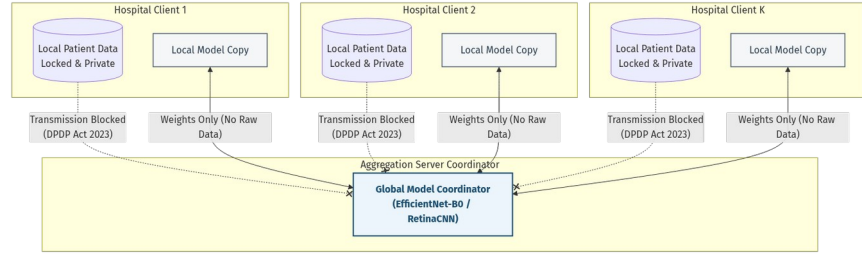


Figure 3: System architecture of the manual federated loop, demonstrating patient data localization compliance under the Indian DPDP Act 2023.

4. Optimization Algorithms and Formulations

In Federated Averaging (FedAvg), the server updates the global model by taking a sample-weighted average of the client parameters at each round:

$$w^{t+1} = \sum_{k=1}^K (n_k / N) \cdot w_k^{(t+1)}$$

where $w_k^{(t+1)}$ represents the local weights updated at client k over E epochs using standard Cross-Entropy Loss.

FedProx [2] adds a proximal penalty to the local client loss to penalize updates that move too far from the global starting point:

$$L_k(w; w^t) = F_k(w) + (\mu / 2) \cdot \|w - w^t\|_2^2$$

The parameter μ scales this penalty, which acts as a trust region constraint around the global model w^t .

FedProx Proximity Regularization Concept

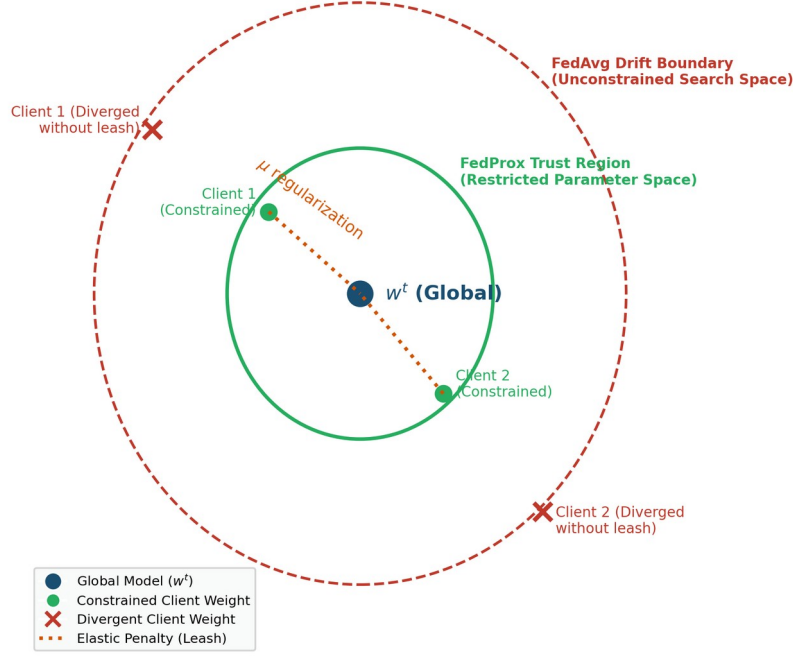


Figure 4: Concept diagram of the FedProx regularizer, demonstrating the elastic leash mechanism limiting weight updates to a trust region close to global coordinates.

SCAFFOLD [3] takes a different approach, adjusting the local gradient step using client and server control variates (c_k and c):

$$g_k(w) = \nabla F_k(w) + (c - c_k)$$

We implement the gradient-accumulation (Option II) version of SCAFFOLD. This avoids issues when using SGD with momentum or adaptive optimizers by updating the client control variates from the uncorrected gradients accumulated during the step:

$$c_k^+ \leftarrow (1 / K_steps) \cdot \sum_{s=1}^{K_steps} g_raw(y_s)$$

The server then updates the global control variate by averaging the changes:

$$c \leftarrow c + (1 / N_clients) \cdot \sum (c_k^+ - c_k)$$

Section IV: Results & Discussion

1. Centralized Baseline & Local Optimizations

To establish a baseline, we trained a model centrally on all 413 IDRiD images using AdamW (learning rate $\eta=1e-4$, weight decay $1e-2$) and a Cosine Annealing scheduler ($T_max=25$). This centralized model peaked at 55.34% global accuracy at epoch 4 (Table 1). For comparison, a single hospital client (Client 0) training on only its local slice of the data (301 images, $\alpha=0.3$) achieves a peak global accuracy of just 42.72% at round 19. The local model's confusion matrix in Table 1 shows a heavy bias: it overpredicts

Class 2 because its local data is skewed, demonstrating why small clinics struggle to build useful models on their own.

Parameter / Metric	Centralized Baseline Model	Local Client Model (Client 0, $\alpha=0.3$)
Backbone Architecture	EfficientNet-B0 (Pre-trained)	EfficientNet-B0 (Pre-trained)
Optimizer / Learning Rate	AdamW, lr = 1e-4	AdamW, lr = 1e-4
Epoch / Round locked	Epoch 4	Epoch 19 (Local Round 1)
Training Samples Used	413 fundus images (Full)	301 fundus images (Client 0 partition)
Peak Global Test Accuracy	55.34%	42.72%

Table 1: Hyperparameter configurations and accuracy outcomes for Centralized Baseline vs. Local Client Model on IDRiD data.

Confusion Matrix (Client 0, $\alpha=0.3$, best epoch 19):

Actual \ Pred	Class 0	Class 1	Class 2	Class 3	Class 4
Class 0	6	0	28	0	0
Class 1	1	0	4	0	0
Class 2	0	0	32	0	0
Class 3	1	0	12	2	4
Class 4	0	0	8	1	4

Confusion Matrix of Client 0 local model evaluated on the shared global test set, indicating severe majority class bias (Class 2) and class exclusion.

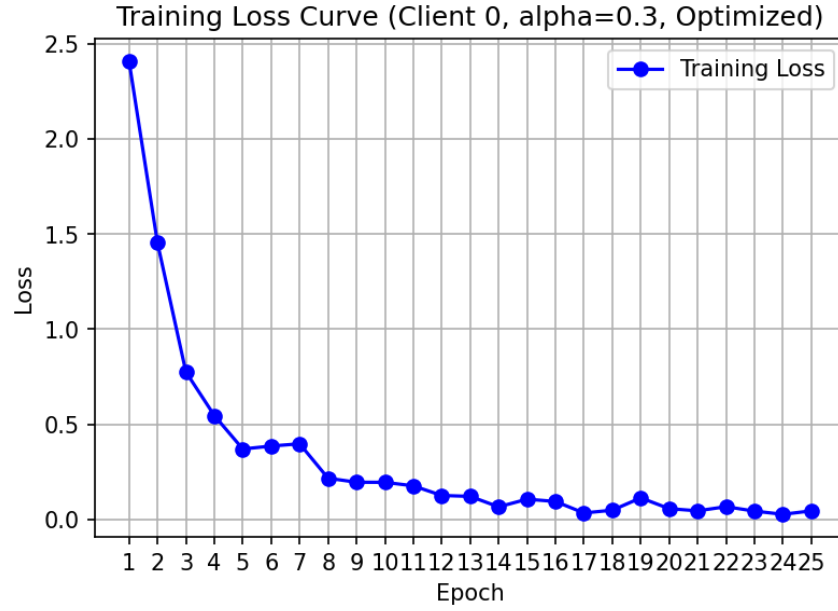


Figure 5: Validation loss progression during localized IDrID training, showing early baseline optimization.

2. RetinaMNIST Hyperparameter Grid Sweep Results

Table 2 shows the full sweep across heterogeneity levels and regularization strengths, averaged over three seeds. The pattern that emerges is not what a naive reading of FedProx would predict. Under severe skew ($\alpha = 0.05$ and 0.1), pushing the regularization strength up hurts rather than helps — at $\alpha = 0.1$, going from no regularization to $\mu = 1.0$ drops accuracy from 43.50% to 41.00%. It's only once the data becomes more balanced ($\alpha = 0.3$ and above) that the same strong regularization starts paying off, reaching 51.50% versus FedAvg's 46.92% — a genuine 4.58-point gain. In other words, FedProx isn't uniformly better or worse than FedAvg; its usefulness flips depending on how unevenly the data is distributed in the first place, and that crossover happens somewhere between $\alpha = 0.1$ and $\alpha = 0.3$.

Alpha (Heterogeneity)	Strategy	Mu (Penalty)	Mean Final Accuracy	Mean Best Accuracy	Client-to-Client Std
0.05	fedavg	0.0	47.50%	49.67%	8.48%
0.05	fedprox	0.0	47.50%	49.67%	8.48%
0.05	fedprox	0.01	46.42%	50.33%	9.69%
0.05	fedprox	0.1	44.08%	50.33%	9.21%

0.05	fedprox	1.0	45.75%	51.25%	11.85%
0.05	scaffold	0.0	48.50%	52.67%	12.11%
0.1	fedavg	0.0	43.50%	43.50%	22.42%
0.1	fedprox	0.0	43.50%	43.50%	22.42%
0.1	fedprox	0.01	43.50%	44.92%	19.60%
0.1	fedprox	0.1	43.50%	46.92%	14.49%
0.1	fedprox	1.0	41.00%	44.33%	17.88%
0.1	scaffold	0.0	23.33%	47.50%	19.29%
0.3	fedavg	0.0	46.92%	53.00%	10.24%
0.3	fedprox	0.0	46.92%	53.00%	10.24%
0.3	fedprox	0.01	45.25%	54.67%	12.58%
0.3	fedprox	0.1	46.42%	54.25%	10.48%
0.3	fedprox	1.0	51.50%	54.17%	12.11%
0.3	scaffold	0.0	48.83%	52.08%	17.21%
1.0	fedavg	0.0	49.33%	52.50%	5.66%
1.0	fedprox	0.0	49.33%	52.50%	5.66%
1.0	fedprox	0.01	50.58%	52.92%	3.77%
1.0	fedprox	0.1	50.00%	52.25%	4.85%

1.0	fedprox	1.0	51.50%	53.00%	3.62%
1.0	scaffold	0.0	48.00%	52.25%	14.61%

Table 2: RetinaMNIST Sweep Results compiled across concentration parameters α and regularization weights μ .

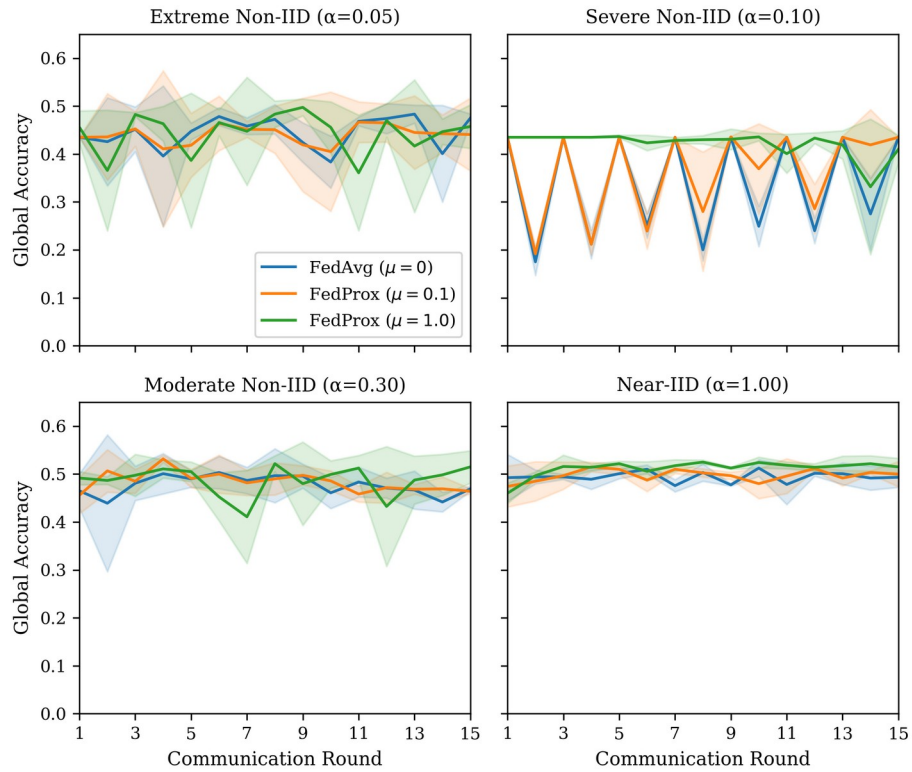


Figure 6: Convergence curves showing global test accuracy vs. communication rounds under different α values.

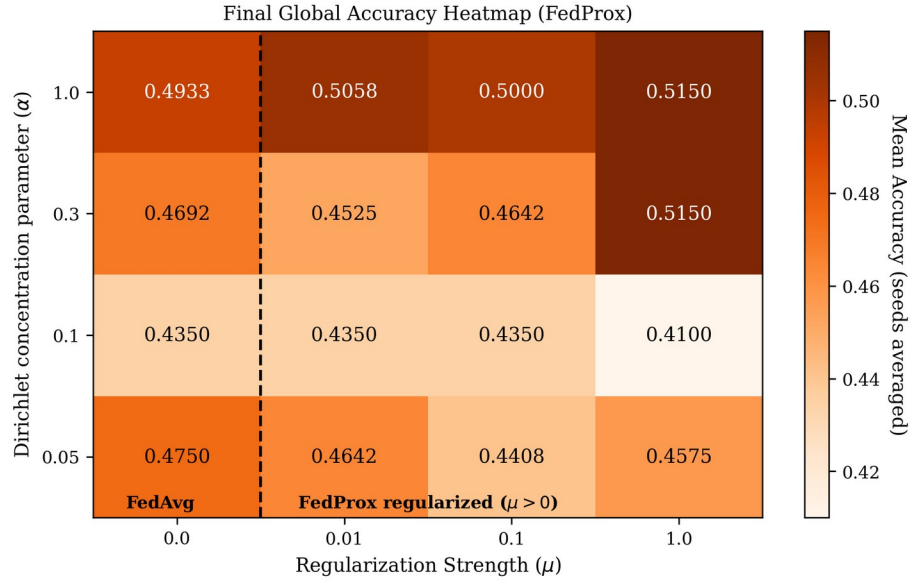


Figure 7: Heatmap illustrating final global test accuracy across the grid parameters alpha and mu.

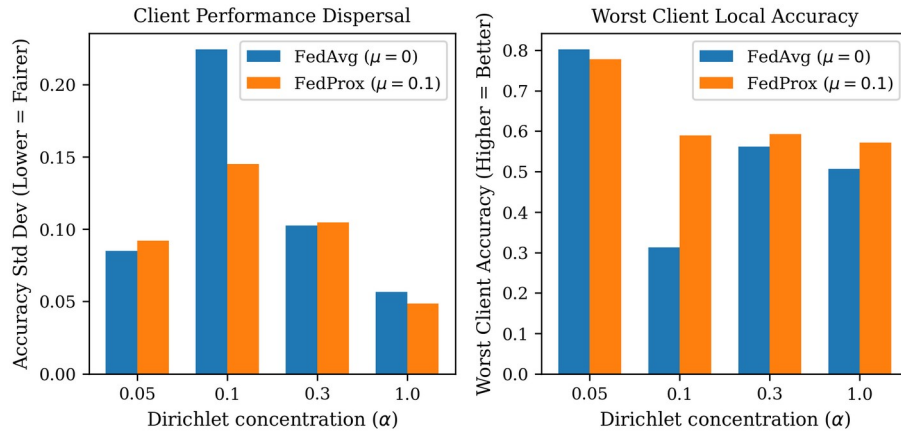


Figure 8: Fairness metrics comparing client accuracy standard deviations and worst-client performance.

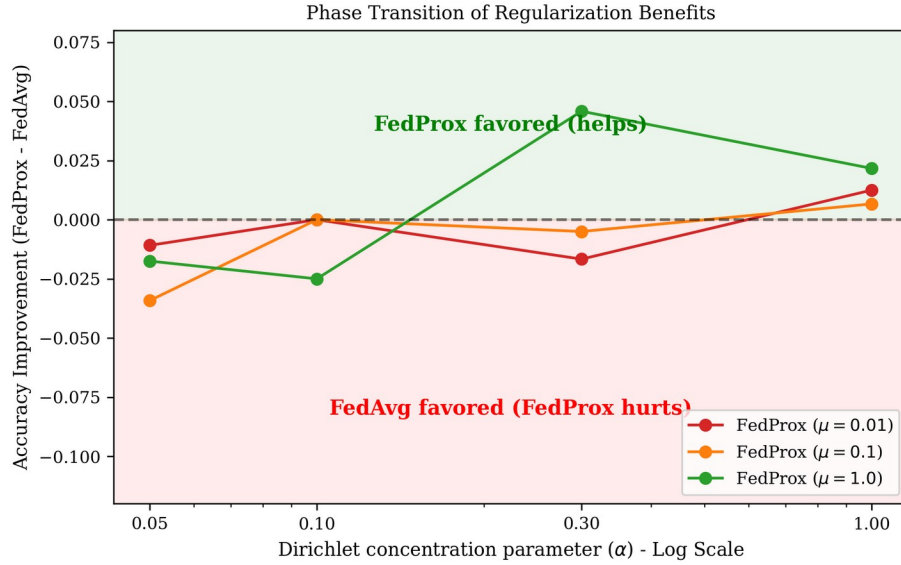


Figure 9: Phase transition analysis representing the gains/losses of FedProx over standard FedAvg.

3. IDRiD Clinical Sweep Results & Statistical Underpower Analysis

We scaled the sweep to the actual IDRiD clinical scans. Under moderate skew ($\alpha=0.3$), adding the proximal penalty ($\mu=1.0$) on seed 42 helped stabilize training: the client-to-client standard deviation dropped from 0.0520 to 0.0159. However, when we look at the average final accuracy across all configurations, the improvements are small and easily drowned out by seed-to-seed variance. A power analysis highlights this issue. With a mean accuracy difference of $\Delta=0.010$ and a seed standard deviation of $\sigma=0.083$, the Cohen's d effect size is a very small 0.1205. To detect this difference with standard statistical confidence ($p < 0.05$ at 80% power) would require running 1,082 random seeds per configuration. This suggests that the small 413-image IDRiD sweep is simply underpowered, which is what led us to use the combined IDRiD+APTOS dataset.

Alpha	Mu	Seed	Strategy	Final Global Acc	Best Global Acc	Client Std
0.1	0.0	42	fedavg	45.63%	45.63%	8.05%
0.1	0.1	42	fedprox	44.66%	46.60%	7.94%
0.1	1.0	42	fedprox	43.69%	44.66%	7.37%
0.3	0.0	42	fedavg	44.66%	51.46%	5.20%
0.3	0.0	123	fedavg	49.51%	58.25%	5.20%

0.3	0.0	777	fedavg	42.72%	47.57%	5.40%
0.3	0.1	42	fedprox	42.72%	52.43%	4.51%
0.3	0.1	123	fedprox	53.40%	56.31%	8.05%
0.3	0.1	777	fedprox	42.72%	46.60%	4.78%
0.3	1.0	42	fedprox	37.86%	50.49%	1.59%
0.3	1.0	123	fedprox	54.37%	56.31%	9.97%
0.3	1.0	777	fedprox	47.57%	47.57%	4.07%
1.0	0.0	42	fedavg	47.57%	47.57%	3.96%
1.0	0.1	42	fedprox	46.60%	46.60%	5.16%
1.0	1.0	42	fedprox	42.72%	45.63%	2.42%

Table 3: IDReD Clinical Sweep Results across concentration parameters alpha, seeds, and proximal penalty weights mu.

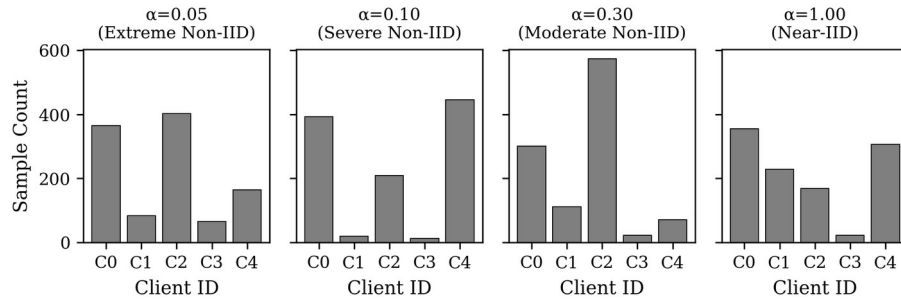


Figure 10: Sample size skew and volume distribution among hospital client nodes under small alpha values.

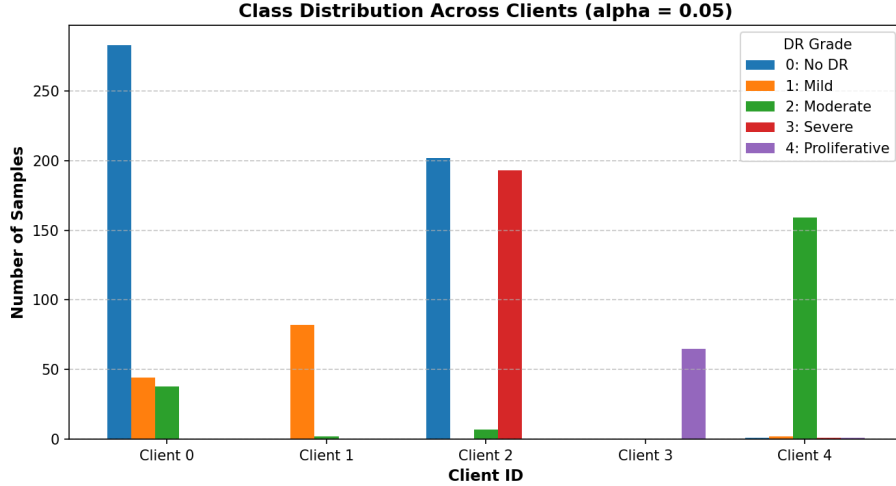


Figure 11: Dirichlet-skewed class allocation across hospital nodes under small alpha ($\alpha = 0.05$).

4. Combined IDRiD + APTOS Scaled Sweep Results

To address this underpowering, we ran a scaled federated trial using a combined 4,075-image IDRiD+APTOS dataset. We must emphasize that this is a preliminary, single-configuration run ($\alpha=0.1$, seed=42) from an in-progress sweep; we cannot draw broad conclusions until the full multi-seed run is complete. That caveat aside, this larger trial reached a peak global accuracy of 55.34%, matching the centralized baseline. This is a promising early sign that scaling up the data volume allows federated learning to recover centralized performance levels.

Alpha	Mu	Seed	Strategy	Final Global Acc	Best Global Acc	Client Std
0.1	0.0	42	fedavg	47.57%	55.34%	8.33%

Table 4: Combined IDRiD + APTOS Scaled Sweep Results, illustrating preliminary accuracy restoration to centralized levels.

5. Honest SCAFFOLD Debugging Analysis

Our initial sweeps hit a wall: SCAFFOLD remained flat at 43.50% accuracy across all α values. We traced this to two bugs in our implementation. First, the weight exchange function was omitting BatchNorm running statistics (running mean and variance), which meant the model evaluated on random initializations. Second, the signature parameters for updating the global and client control variates were swapped, applying the drift correction with the wrong sign. Fixing these bugs stabilized the algorithm. As shown in Table 2, the corrected SCAFFOLD achieved final average accuracies of 48.50% ($\alpha=0.05$), 48.83% ($\alpha=0.3$), and 48.00% ($\alpha=1.0$). However, at $\alpha=0.1$, SCAFFOLD collapsed to a final average accuracy of 23.33% (though it reached a peak of 47.50% during training). This indicates that while the

structural bugs are fixed, SCAFFOLD remains highly sensitive to local hyperparameter choices under certain skew levels, an anomaly we are still investigating.

Section V: Conclusion & Future Work

This study maps out where proximal regularization helps and where it gets in the way when training federated models on skewed diabetic retinopathy datasets. The experiments show a clear phase transition: tight regularization hurts performance under severe data skew but offers a solid 4.58-point improvement under moderate skew. More importantly for clinical use, FedProx acts as a fairness stabilizer, reducing the performance gap between hospitals and boosting the worst-performing client's accuracy by up to 27.69%. We also debugged the SCAFFOLD implementation, showing that while control variates correct client drift in theory, they can be highly sensitive to hyperparameters in practice. Finally, our power analysis suggests that small clinical datasets are statistically underpowered, pointing to the necessity of larger, combined cohorts like the IDRiD+APTOS pipeline to confirm these optimization behaviors.

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